



Propagation of Angular Distribution Uncertainties in Vessel Fluence Calculations

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ABSTRACT

Accurate estimation of fast neutron fluence at the reactor pressure vessel (RPV) is essential for structural-integrity assessments and long-term operation of pressurized water reactors (PWRs). While uncertainties in reaction cross sections have been extensively studied, significantly less attention has been given to the uncertainties in elastic-scattering angular distributions (AD), despite their key role in deep-penetration problem. This work quantifies the impact of AD variability on vessel-fluence predictions using the surveillance capsules of Turkey Point Unit 3 (TP3) as a realistic application case.

A coupled Polaris–GenPMAXS–PARCS–Serpent2 workflow was established to propagate uncertainties from core-follow calculations to ex-core shielding analysis, and three modern nuclear data libraries (JEFF-4, JENDL-5, and ENDF/B-VII.1) were assessed against publicly available activation measurements.

Because MF=34 covariances for scattering anisotropy cannot yet be processed within EPFL's current toolchain, an ad-hoc approach was developed. Samples of perturbed angular distributions were generated using TALYS/T6, and the resulting MF=4 records were substituted into the nominal file while keeping the MF=3 cross sections fixed. The spread of these perturbed ADs was verified to encompass relevant EXFOR differential-scattering data, at least at high energies. Propagation of the sampled ADs through the full TP3 model indicates an uncertainty contribution of approximately 4% at capsule locations—smaller than the dominant nuclear-data and geometric contributions, except for the JEFF-4 library where the uncertainties are much smaller, below 1%.

Keywords: Nuclear Data, Angular Distribution, PWR, Vessel Fluence, Long Term Operation

1. INTRODUCTION

The accurate estimation of fast neutron fluence at the reactor pressure vessel (RPV) is central to reactor lifetime extension and structural integrity assessments in pressurized water reactors (PWRs). As demonstrated in recent studies within the EVEREST project [1], uncertainties in neutron transport modeling, especially those arising from nuclear data, remain among the main contributors to the overall uncertainty in vessel fluence predictions [2, 3, 4]. Conventional analyses have largely focused on reaction cross-section uncertainties, while less attention has been devoted to the angular distribution (AD) of secondary neutrons resulting from elastic and inelastic scattering collisions. These distributions play a crucial role in deep-penetration and attenuation-dominated transport, where forward scattering determines the spatial attenuation of the fast flux through structural materials. This is mainly due to historical limitations in covariance evaluations and data-processing capabilities. Modern general-purpose

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nuclear data libraries, such as JEFF, JENDL, and ENDF/B, now include more complete covariance information, enabling the propagation of AD uncertainties within vessel fluence calculations.

The present work pursues two main objectives. First, it assesses the performance of three general-purpose nuclear data libraries (JEFF-4, JENDL-5, and ENDF/B-VII.1), in predicting fast neutron fluence at capsule locations during various cycles of a PWR for which public data is available. Second, it investigates an ad-hoc approach to evaluate the contribution of scattering angular-distribution uncertainties to the total fluence uncertainty. To date, EPFL does not possess the infrastructure or tools to generate perturbed MF=4 (angular-distribution) samples directly from the corresponding MF=34 covariance data. This limitation motivated the development of an alternative route based on the TALYS nuclear reaction code[5], which can produce consistent sets of optical-model-derived angular distributions and associated covariances. The present work proposes and tests a methodology to approximate the effect of angular-distribution variability within vessel fluence assessment calculations. The resulting framework complements another paper also submitted to PHYSOR 2026 [6] by providing an alternative path to include AD effects in uncertainty propagation; and its application beyond simple benchmarks or experiments to a full PWR fluence analysis.

2. CASE STUDY: THE CAPSULES S AND T OF TURKEY POINT 3 NUCLEAR POWER PLANT

The Turkey Point 3 (TP3) reactor is a Westinghouse three-loop PWR with a nominal power of 2200 MWth (693 MWe). The reactor exceeded 50 years of operation with an average load factor of 80% and has received a subsequent license renewal to operate until 2052. The reactor physics and core operating data of the first three cycles of TP3 have been the subject of recent validation efforts for the SCALE/Polaris-PARCS code procedure [7]. Public reports detail core loading maps, power histories and relevant ex-core dimensions for shielding problems. Surveillance capsules are attached to the reactor thermal shield to monitor material damage during irradiation. Derived fluence from capsule activation measurements is presented in public Westinghouse reports, completing the set of information necessary to set-up a preliminary validation case for a vessel fluence calculation methodology. Figure 1 displays the main features of the TP3 model for fluence calculations, highlighting the position of capsules T and S, which will be the object of the uncertainty investigation presented in this paper.

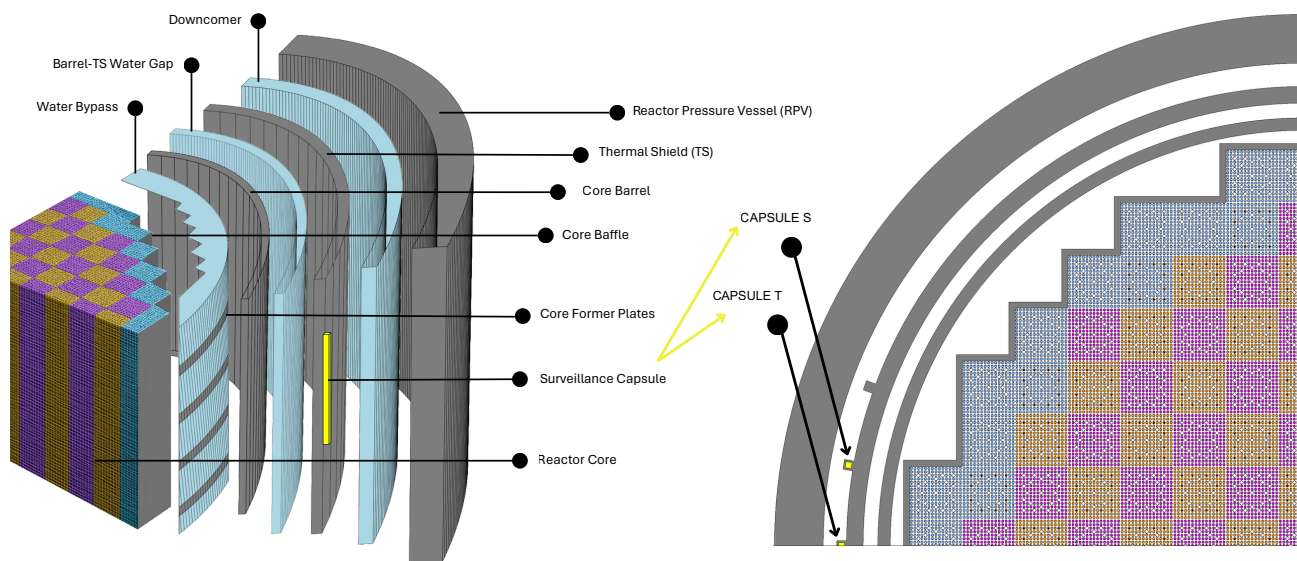


Figure 1. Turkey Point 3 vessel fluence model and target quantities.

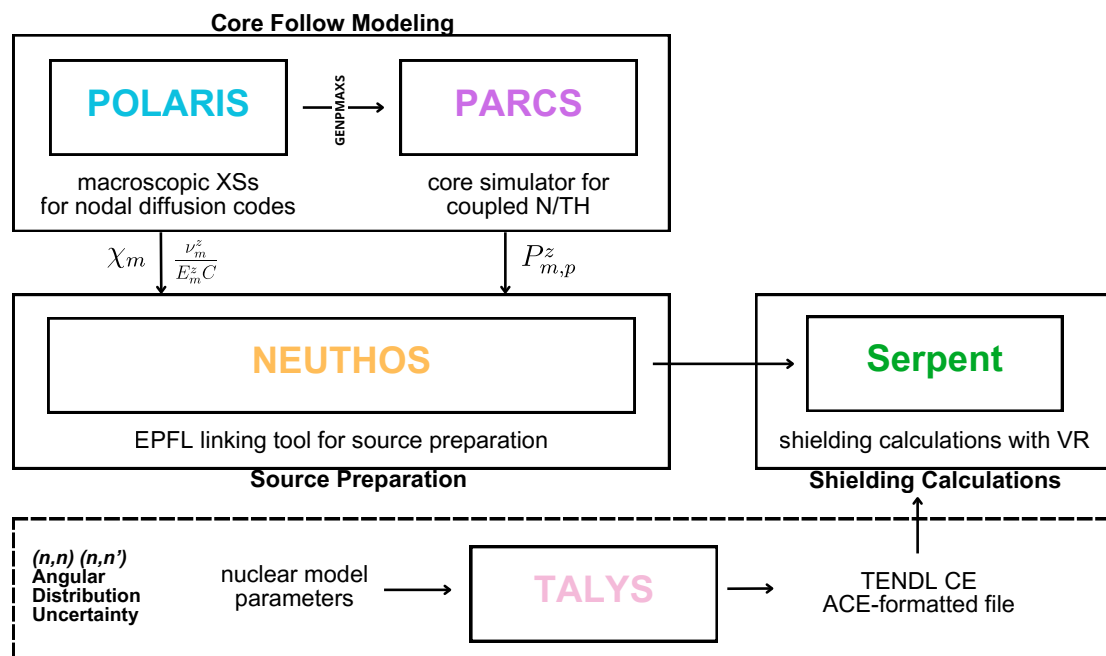


Figure 2. Capsule fluence estimation methodology.

3. METHODOLOGY

This section outlines the approach involved in this work, from determining the neutron fluence at the capsule location to propagating the AD uncertainties through the computational model.

3.1. Vessel fluence calculations

The developed methodology at EPFL[8] couples deterministic and stochastic transport tools to evaluate neutron fluence at various capsule locations. Core-follow calculations are performed using the Polaris–GenPMAXS–PARCS code suite, where Polaris generates two-group cross-section libraries and pin-power factors. GenPMAXS processes the cross-section tables for PARCS, which executes nodal diffusion simulations with internal thermal–hydraulic feedback. Pin-wise power data across the axial mesh are extracted and converted into local neutron production by fission by means of a Python-based linking tool (NEUTHOS[2]) that retrieves fission rates, nuclide densities, and spectral data from the lattice physics results.

Then fixed sources Monte Carlo simulations are performed with Serpent2. The fission spectrum is specified from literature-based Watt and Maxwellian functions. Serpent2 uses these sources in fixed-source mode with response-matrix-based weight-window variance reduction techniques to compute the fast neutron flux at the surveillance capsule. This workflow is depicted in Figure 2. It enables consistent propagation of modeling assumptions and input uncertainties from the core to ex-core regions.

3.2. Uncertainty propagation

In a previous paper [2], various UQ techniques (stochastic sampling and sandwich formula) have been combined to estimate computational uncertainties associated with neutron fluence at the capsule locations.

The list of uncertain parameters considered in [2] is:

- nuclear data in core follow calculations;
- nuclear data in shielding calculations;
- source definition (fission spectrum and neutron sampling probability);
- in-core geometry specifications (fuel, cladding and grid spacers);
- ex-core geometry specifications including core baffle, core barrel, thermal shield and reactor pressure vessel;
- isotopic composition and material densities, considering structures and coolant in both core follow calculations and shielding calculations;

A summary of the results described in [2] is reproduced in Table I. The various components of the computational uncertainty can be split by nuclear data (ND) and technology parameters (TP). In core-follow, both TP (here, thermal-hydraulic boundary conditions) and ND perturbations have a negligible impact on the capsules fluence. By contrast, in shielding calculations both ND and TP are relevant. The perturbation of AD is not included in the nuclear data uncertainties reported in Table I.

Table I. Magnitude of UQ components (fast flux $E > 1$ MeV) at the capsule locations.

Component	Uncertainty
<i>Core-follow (source definition)</i>	
TP (TH boundary conditions)	0.17% $\pm$ 0.04
ND (MF=33)	0.26% $\pm$ 0.06
<i>Shielding (fixed-source transport)</i>	
TP (manufacturing tolerances)	5.28%
ND (MF=33)	7.34% $\pm$ 0.53
Total uncertainty	9.94% $\pm$ 0.40

In attenuation-dominated transport problems, such as the neutron transport from the reactor core to the pressure vessel, scattering anisotropy and more specifically the AD have a large impact on how neutrons traverse structural materials between the core and the vessel. The following section details the methodology adopted to incorporate AD uncertainties associated with Fe-56 in the fluence estimates at capsule locations.

3.3. Angular distributions

3.3.1. Representation of Angular Distributions

In evaluated nuclear data files, elastic-scattering angular distributions are expressed through a Legendre polynomial expansion that characterizes the probability density of scattering angles as a function of energy. The angular probability density function $f(\mu, E)$, with $\mu = \cos \theta$ denoting the cosine of the scattering angle, can be written as:

$$f(\mu, E) = \sum_{\ell=0}^{L_{\max}} \frac{2\ell+1}{2} a_{\ell}(E) P_{\ell}(\mu), \quad (1)$$

where $P_{\ell}(\mu)$ are the Legendre polynomials and $a_{\ell}(E)$ are the energy-dependent Legendre coefficients that define the anisotropy of the scattering process. In this formulation, $a_0(E)$ corresponds to the isotropic term and is normalized to unity, while $a_1(E)$ represents the first-order anisotropic contribution, equal to the mean scattering cosine $\bar{\mu}(E)$. Higher-order terms refine the description of forward and backward scattering components. These coefficients are tabulated in MF=4 of the ENDF-6 format, and their associated covariance matrices are provided in MF=34.

Since the application case of this work is the simulation of vessel fluence for neutron energies larger than 1 MeV, only the AD of elastic scattering are considered. It is assumed that inelastic scattering collisions would slow down the neutrons below 1 MeV and as such makes the AD irrelevant for the estimation of the vessel fluence mean and uncertainty (the inelastic cross section itself still affects the neutron transport).

3.3.2. Sampling Strategy and Data Substitution

Even though angular distribution covariances are available in the Fe-56 file of the nuclear data libraries considered in this work, EPFL does not possess the infrastructure or tools to generate perturbed MF=4 (AD) samples directly from the corresponding MF=34 covariance data. A work around has been designed for the purpose, relying on the T6 suite of codes. It is described in the present section, focusing mainly on the ENDF/B-VII.1 library without loss of generality.

The underlying optical model parameters of TALYS[5] for Fe-56 are randomly sampled 100 times from default uniform distributions available in TALYS; and TALYS is rerun for each perturbed parameter set. The resulting ensembles of cross sections and angular distributions are then formatted as perturbed ENDF files, preserving the correlations between MF=3 and MF=4 quantities.

Then, the MF=4 (elastic) record of the nominal TALYS Fe-56 file is replaced by perturbed records MF=4 (elastic) obtained previously. The elastic scattering cross section (MF=3) is kept at the nominal value of TALYS; this isolates the uncertainties associated with AD but it is not self-consistent. Indeed ENDF/B-VII.1 and TALYS elastic cross sections do not overlap as it is visible in Figure 3(a). Yet the general trend of the cross sections with energy are very similar and as a result, the proposed approach should lead to reasonable uncertainty estimates (the nominal computational scheme relies on the ENDF/B-VII.1 file for Fe-56). This inconsistency is the reason why this method is considered ad-hoc.

The a_1 relative uncertainties evolution with energy together with the correlation matrix is shown in Figure 3b and Figure 3c respectively. It should be noted here that the correlations produced this method are consistent with the equivalent information of JENDL-5 reported in [6].

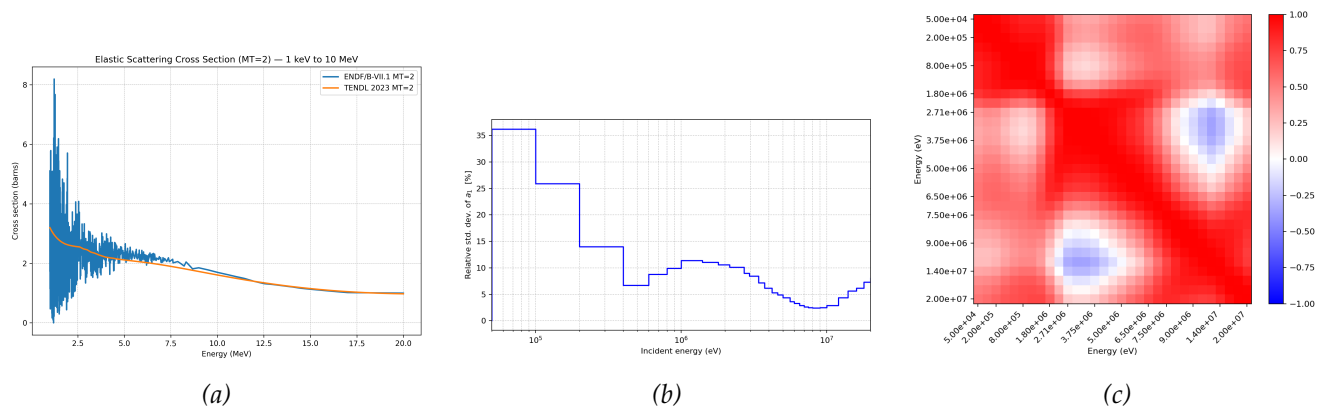


Figure 3. (a) elastic scattering cross section comparison between ENDF/B-VII.1 and TALYS; (b) relative standard deviations of a_1 (c) a_1 correlation matrix.

Next, the spread of the Fe-56 differential scattering cross sections in the 100 random files are compared with a subset of available EXFOR differential elastic scattering data for Fe-56. The results are shown in Figure 4 for various incoming neutron energies. The plotted quantity, $d\sigma/d\Omega$, corresponds to the differential cross section as a function of the scattering angle. The energy resolution of the measurement has not been taken into account, which might affect the comparison between the files and the EXFOR data below 2 MeV.

The spread of these TALYS-derived differential cross sections generally encompasses the experimental measurements at all angles for high energies (above 2 MeV). This is a strong indication that the TALYS parameter-space perturbations generate uncertainties of the correct magnitude. However, at lower energies, the spread of the random differential cross sections does not cover the EXFOR data which can be attributed to the energy resolution of the measurement not being taken into account.

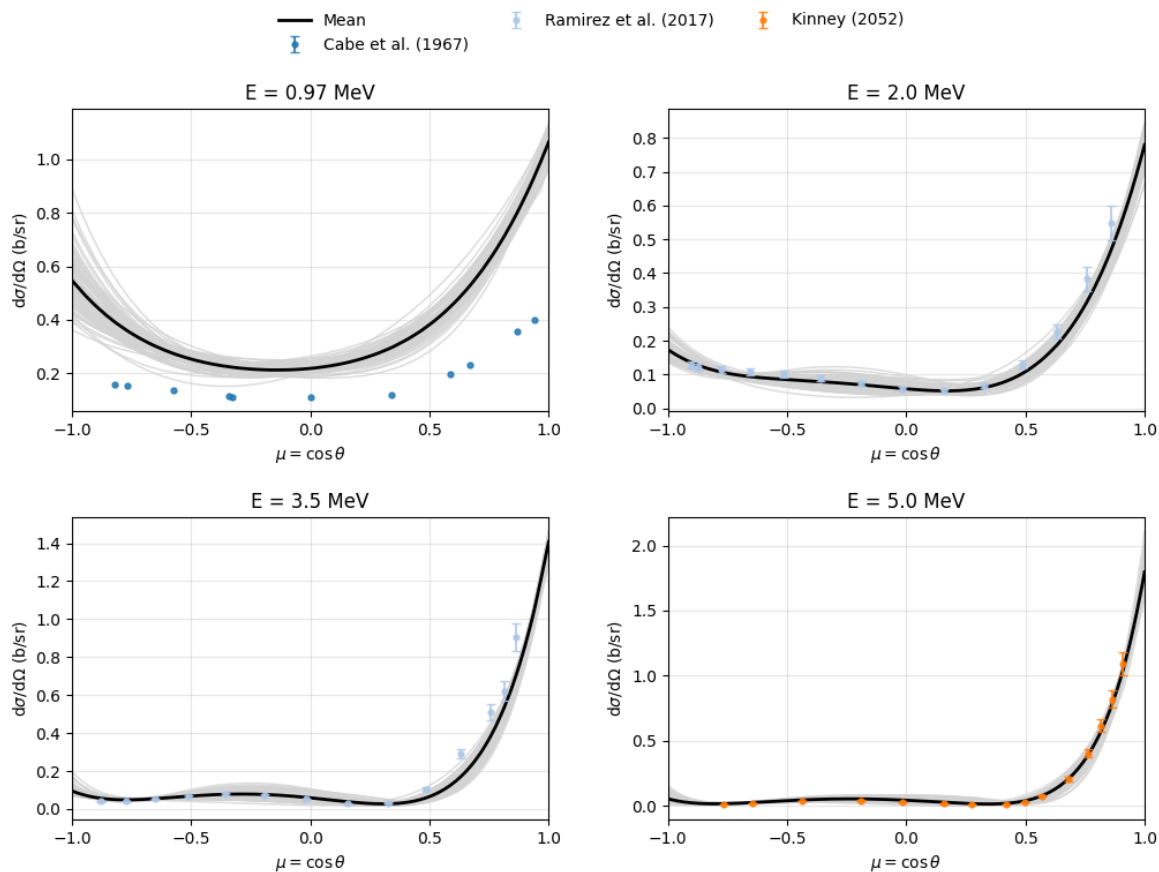


Figure 4. Fe-56 elastic angular distributions compared with EXFOR data for various energies. Black: mean of 100 samples; light grey: each sample.

4. RESULTS

Next the performance of three general-purpose nuclear data libraries (JEFF-4, JENDL-5, and ENDF/B-VII.1) is assessed with respect to the prediction of fast neutron fluence at capsule locations after the first (capsule T) and fourth cycle (capsule S) of the TP3 nuclear power plant. In this part we focus on the mean value and the uncertainties due to AD only.

The method described in Section 3.3.2 is used to obtain the AD uncertainties of ENDF/B-VII.1. For JEFF-4 and JENDL-5, random samples of the Fe-56 AD generated within the work reported in [6] are employed.

Due to time constraints, only a very limited number of random samples (50) could be considered. It is enough for a coarse grain estimate ($\approx 10\%$ uncertainty in the standard deviation) of the AD uncertainty. These results will be updated at the conference when more fluence samples become available.

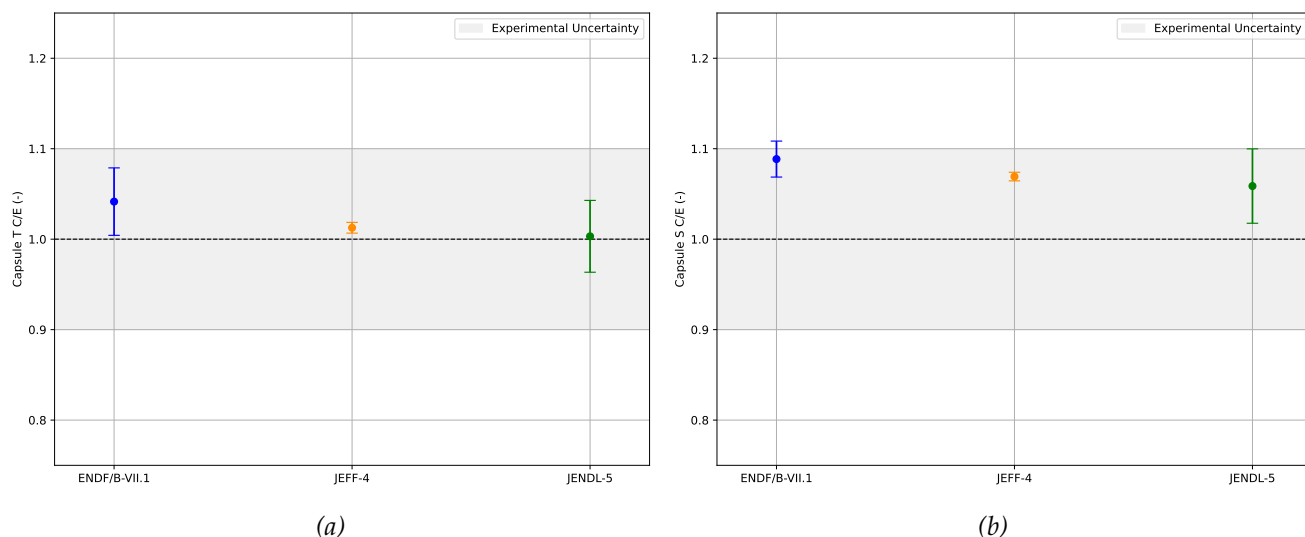


Figure 5. Comparison calculations vs measurements (C/E) for various nuclear data libraries and estimates of the associated computational uncertainties (1σ) due to AD; for (a) capsule T; (b) capsule S.

Figure 5 shows the comparison of calculated and measured fast neutron fluence at the surveillance capsule locations using JEFF-4, JENDL-5, and ENDF/B-VII.1; after cycles 1 and 4 of TP3. Across both capsule locations, the nominal predictions from all three libraries remain well within the experimental uncertainty (10%), with JENDL-5 generally providing the closest agreement in both cycles. ENDF/B-VII.1 and JEFF-4 tend to yield slightly higher fluence estimates. Due to the large experimental uncertainties, such measurements are not really conclusive to judge on the better or worse performance of a given nuclear data library.

The uncertainty bands shown in Figure 5 correspond solely to the variability of the elastic-scattering AD for Fe-56. Their magnitudes vary widely from one library to the next as also reported in [6] but for simpler problems. AD uncertainties results in fluence uncertainties at both capsule locations in the 2–4% range for TALYS-based ENDF/B-VII.1 and JENDL. They are noticeably smaller than the dominant components listed in Table I. As for JEFF-4, the capsule uncertainties are significantly smaller; specifically, the AD uncertainties (between 0.3 and 0.6%) are in the range of the statistical error of these shielding calculations (between 0.2 and 0.3%).

5. CONCLUSIONS

This work investigated the role of elastic scattering AD uncertainties in RPV fluence estimation for four cycles of an operated PWR, the Turkey Point Unit 3 power plant. A complete core-follow and shielding calculation set based on the Polaris–GenPMAXS–PARCS–Serpent2 methodology was employed to generate fluence estimates at capsule locations, using three modern general-purpose nuclear data libraries (JEFF-4, JENDL-5, and ENDF/B-VII.1). The computational results are compared to the same quantity derived from activation measurements.

Because EPFL currently lacks the capability to process MF=34 angular-distribution covariances into perturbed MF=4 files, an alternative pathway was implemented using a TMC approach relying on the T6 codes. Except at energies below 2 MeV, the TALYS-generated random AD samples were shown to span the range of available EXFOR differential elastic-scattering data, despite the fact that these measurements were not included in the T6 parameter optimization. This supports the fact that AD uncertainties produced by T6 are realistic. This approach was used to estimate the AD uncertainties for ENDF/B-VII.1, while preexisting samples were used for the JEFF-4 and JENDL-5 libraries.

In terms of fluence calculations, for JENDL and ENDF/B-VII.1, the propagated AD induced uncertainties at the capsule locations were found to be on the order of 2–4%, smaller than other uncertainty contributors. For JEFF-4, the AD uncertainties are below 1%, e.g. same range of the statistical error obtained in these shielding calculations. Such result is surprising and will be further investigated in more details.

Overall, the work demonstrate that (i) all three evaluated libraries reproduce the measured capsule fluence within experimental uncertainties, (ii) the ad-hoc TALYS-based approach provides a reasonable estimate of AD uncertainties.

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